

Santosh Tirumala

730 E Evelyn Avenue Apt 221, Sunnyvale CA 94086

<https://skbt27i.netlify.app/> ❖ <https://github.com/skbt27i>

skbt27i@gmail.com ❖ 813-414-1642

EDUCATION

University of Florida, Gainesville

August 2018- May 2022

B.S., Computer Science

Minor: Mathematics

GPA: 3.56 / 4.00

EXPERIENCE

Amazon Lab126

July 2022 – Present

Software Development Engineer 2

- Worked on Speech/Audio AI Benchmarking team, which evaluates speech/audio output on various services and devices
- Onboarded several metrics, including 2 Airflow DAGs automating previously manual processes, for testing speech-to-speech foundation model capabilities, exposing areas for improvement in the process, including fundamental model configuration changes. **Enabled 20%+ improvement in a model metric for an external leaderboard**
- Implemented smart TV benchmarking framework; allowed manipulation of smart TV remote, and included computer vision capabilities. Technologies used include home robotics, Raspberry Pi, Arduino, capture cards, AWS Rekognition, AWS Batch, AWS Step Functions, AWS Lambda, AWS DynamoDB, AWS S3, AWS CloudWatch, AWS IAM, Python, Java
- Implemented mobile automation framework, enabling full control of smartphones. Features included opening/closing apps, gracefully handling popups or crashes, and pressing buttons, in a fully extensible manner. Technologies used include AWS Lambda, AWS Step Functions, AWS DynamoDB, AWS S3, AWS CloudWatch, AWS IAM, Appium, UIAutomator, Python, Java. Saved **\$200k** on manual efforts
- Implemented scalable API benchmarking system, with features like multithreading and request scheduling. Technologies used include AWS ECS (Fargate), AWS Batch, AWS SQS, AWS CDK, AWS Step Functions, AWS DynamoDB, AWS S3, AWS CloudWatch, AWS API Gateway, AWS IAM, Python, TypeScript
- Maintained, refactored and improved JavaScript+React+Node centralized portal for the team's services. Also worked as away team on AI org's centralized UI (TypeScript+React+Node), replacing deprecated leaderboard solution
- Led operational improvements such as unit/integration testing coverage, Claude Code skills for automating routine tasks, and improved testing environments in several team/away team services
- Guided 3 interns and 2 full time hires, onboarding them to the team's services
- Provided technical assistance to internal customers on the team's services

UNIVERSITY COURSES

Artificial Intelligence, Computer Graphics, Software Engineering, Data Structures, Computer Organization, Digital Logic, Performant Python, Operating Systems, Applied Machine Learning

SKILLS/TECHNOLOGIES

Knowledge of AWS services, Agile Scrum, Linux, OpenGL, Node.js, React, SQL, Git, Docker, Kubernetes, PyTorch, Claude Code CLI, Amazon Kiro CLI

Best in Java and Python; Knowledge of JavaScript+HTML+CSS, TypeScript, C++